Determinantal Harmonic Resonance

Gauss–Vandermonde finite parts, Stieltjes jets, and depth-two polylogarithmic closure

A research continuation for the ProveIt project

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Abstract. We develop a determinantal extension of the Gamma–harmonic identity programme. A self-contained evaluation of a discrete Gauss–Vandermonde sum is used at its first logarithmic resonance. Separating the largest summation index produces a normally convergent subtraction, an explicit hard-cutoff constant, and all-order identities coupling many-variable harmonic sums to generalized Stieltjes constants. The same construction has exact Hurwitz-zeta derivatives and anchored log-Gamma primitives.

For rational weights with simple poles on the negative integer grid, we prove that the finite moment matrix has harmonic dependence in only one entry. Consequently an arbitrary number of summation variables still admits a depth-two polylogarithmic generating function. Every positive Euler primitive at the endpoint reduces to ordinary zeta values and products of two zeta values. We give explicit two-, three-, and four-variable evaluations, an all-order mixed harmonic generating function, and a common-translate cyclotomic extension.

The proofs, exact rational reducers, and diagnostic calculations are separate parts of the delivery. The replay contains 732 exact assertions and 85 numerical diagnostics at 100-digit working precision. The latter are not interval certificates. We do not assert global priority, resolve the manuscript's S_6 or S_8 conjectures, or confer canonical or proof-assistant status on this draft.

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Scope and integration qualifications are recorded in Section 8.

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1 Context, scope, and conventions

1.1 The continuation problem

The canonical `ProveIt` manuscript already has a substantial Gamma–harmonic and Hurwitz–Stieltjes calculus. In particular, the inspected mixed-Dougall chapter proves a normally convergent two-parameter Gamma germ, all its fixed mixed derivatives, a weighted trigamma square, and a quadratic zeta-tail family [1, 2]. The manuscript also records an all-order Gauss proof of the corrected pure and mixed Choi harmonic families. Re-proving the one-variable coefficient engine would not by itself be a meaningful continuation.

We instead ask what happens when a squared Vandermonde couples an arbitrary number of summation indices. There are two distinct questions. First, can the Gamma product still supply a correctly normalized Stieltjes subtraction when the sum ceases to converge? Second, does the increase in the number of indices force a comparable increase in polylogarithmic depth? We answer both for the first resonant determinantal family.

The first answer requires attention to a distinction easily hidden by formal Laurent expansion: a spectral finite part is generally *not* the constant term of a hard cutoff. The residue amplitude varies with the spectral parameter. Its derivative is precisely the discrepancy. The second answer is more economical than the number of variables suggests: for simple-pole rational weights at critical decay, only the bottom-right moment contains a harmonic number. This gives depth at most two at every particle number.

1.2 Main results and what they do not claim

Theorem 2.1 evaluates the convergent discrete partition function by elementary determinant algebra and Gauss summation. Its classical ingredients are not presented as new. Theorems 3.1 and 4.1 give the normally convergent resonance calculus and its all-order Stieltjes consequences. Theorems 6.1 and 6.3 give the finite-cutoff rank-one harmonic theorem and its all-particle depth-two consequence. The concrete formulas in Section 7 are exact applications, not identities inferred from a numerical relation search.

These are results proved in the present report. An exhaustive literature comparison has not established their global novelty. Nor do they prove a lower bound on the depth of any actual period. The manuscript explicitly distinguishes formal cyclotomic quotients from numerical period independence [3]; we retain that distinction. The short S_6 and S_8 identities remain outside the results proved here.

1.3 Notation and analytic conventions

For $k \geq 0$, write $(x)_k = \Gamma(x+k)/\Gamma(x)$ for a rising factorial, and $x^{\underline{k}} = x(x-1)\cdots(x-k+1)$ for a falling factorial. Empty products and empty determinants are one. We use

$$H_n^{(s)} = \sum_{j=1}^{n} j^{-s}, \qquad H_n = H_n^{(1)}, \qquad H_0^{(s)} = 0.$$

The gamma logarithmic derivative is $\psi = \Gamma'/\Gamma$. Our Stieltjes convention is

$$\zeta(1+u,\alpha) = \frac{1}{u} + \sum_{k\geq 0} \frac{(-1)^k}{k!}\gamma_k(\alpha)u^k, \qquad \operatorname{Re}\alpha > 0. \tag{1.1}$$

Thus $\gamma_0(\alpha) = -\psi(\alpha)$ and $\gamma_0(1) = \gamma$. Derivatives $\zeta^{(k)}(s, \alpha)$ are in the first, or order, variable unless specified otherwise. We use the standard branches of Gamma and Hurwitz zeta on the stated right half-planes [6, 9, 8].

Within $|z| < 1$, define

$$\mathrm{Li}_s(z) = \sum_{n\geq 1} \frac{z^n}{n^s}, \qquad \mathrm{Li}_{p,1}(z,1) = \sum_{n>m\geq 1} \frac{z^n}{n^p m}. \tag{1.2}$$

The latter convention agrees with the two-argument convention $\mathrm{Li}_{p,1}(x,y) = \sum_{n>m\geq 1} x^n y^m/(n^p m)$. In particular $\mathrm{Li}_{1,1}(z,1) = \frac{1}{2}\log^2(1-z)$. All formulas initially in the disk use the logarithm analytic there and zero at $z = 0$. Rational Laurent factors in z in the displayed reductions have removable singularities after the full expression is assembled. Analytic continuation means continuation of this common germ, not an independent choice of branch for each summand.

The complete exponential Bell polynomials are normalized by

$$\exp\left(\sum_{j\geq 1} x_j \frac{t^j}{j!}\right) = \sum_{k\geq 0} \mathcal{B}_k(x_1, \ldots, x_k)\frac{t^k}{k!}. \tag{1.3}$$

For example, $\mathcal{B}_1 = x_1$, $\mathcal{B}_2 = x_1^2 + x_2$, and $\mathcal{B}_3 = x_1^3 + 3x_1x_2 + x_3$.

Whenever a series of subtracted brackets is asserted to converge, the brackets are the summands. This does not license splitting the series into separately divergent components. All index and parameter derivatives in the results below are fixed finite orders.

2 A discrete Gauss–Vandermonde product

For $r \geq 1$ and $0 \leq n_1 < \cdots < n_r$, put

$$\Delta(\boldsymbol{n}) = \prod_{1\leq i<j\leq r} (n_j - n_i), \qquad w_n(a,c;b) = \frac{\Gamma(n+a)\Gamma(n+c)}{\Gamma(n+b)\Gamma(n+1)}.$$

Define

$$Z_r(a,c;b) = \sum_{0\leq n_1<\cdots<n_r} \Delta(\boldsymbol{n})^2 \prod_{i=1}^{r} w_{n_i}(a,c;b), \qquad Z_0 = 1. \tag{2.1}$$

The ordering is part of the definition. There is no additional $r!$ in any product formula below.

Theorem 2.1 (Gauss–Vandermonde product)**.** *Let* $\operatorname{Re} a, \operatorname{Re} c > 0$ *and* $\operatorname{Re} d > 2r - 2$*, where* $d = b - a - c$*. Then* (2.1) *converges absolutely and locally normally, and*

$$\boxed{Z_r(a,c;b) = \prod_{j=0}^{r-1} \frac{j!\,\Gamma(a+j)\Gamma(c+j)\Gamma(d-r+1-j)}{\Gamma(b-a-j)\Gamma(b-c-j)}.} \tag{2.2}$$

For positive real parameters in this domain all summands are positive.

We give the determinant argument in detail, both to make the normalizing factor checkable and to avoid relying on an unproved discrete-Selberg analogy. Gamma ratio estimates and Gauss summation are classical [6, 5]; the determinant technique belongs to the standard evaluation toolkit [11].

Lemma 2.2 (A rational factorial determinant)**.** *For indeterminates $x_0, \dots, x_{r-1}, t$,*

$$\det_{0\le i,j<r} \frac{(x_i)_j}{(x_i+t)_j} = \frac{\displaystyle\prod_{j=0}^{r-1}(t)_j \prod_{0\le i<k<r}(x_k-x_i)}{\displaystyle\prod_{i=0}^{r-1}(x_i+t)_{r-1}}. \tag{2.3}$$

The identity holds at every specialization where the displayed rational functions are defined, and elsewhere by cancellation.

Proof. Multiply row i by $(x_i+t)_{r-1}$. The resulting column polynomials are

$$p_j(x) = (x)_j(x+t+j)_{r-1-j}, \qquad 0 \le j < r.$$

They have degree at most $r-1$, so their evaluation determinant is the Vandermonde in the x_i times their coefficient determinant. To compute that coefficient determinant, evaluate at $x_i = -i$. The evaluation matrix is lower triangular, since $(-i)_j = 0$ for $j > i$, with diagonal

$$p_i(-i) = (-1)^i i!\,(t)_{r-1-i}.$$

Its determinant divided by $\prod_{i<k}((-k)-(-i)) = (-1)^{r(r-1)/2}\prod_{k=0}^{r-1} k!$ is $\prod_{j=0}^{r-1}(t)_j$. Dividing the row factors proves (2.3) as a rational identity. □

Proof of Theorem 2.1. On compact subsets of the stated domain, $w_n = O((n+1)^{-1-\operatorname{Re} d})$. For ordered nonnegative indices,

$$|\Delta(\boldsymbol{n})|^2 \le \prod_{j=1}^{r}(n_j+1)^{2(j-1)}.$$

Dropping the ordering gives a product of convergent majorant series when $\operatorname{Re} d > 2r-2$. This proves absolute local normal convergence, including that of moments of degrees at most $2r-2$.

Apply finite Cauchy–Binet to the two monic polynomial bases $n^{\underline{i}}$ and $(n+a)_j$, then pass to the normally convergent limit. Both evaluation determinants equal the Vandermonde, so

$$Z_r = \det_{0\le i,j<r} M_{ij}, \qquad M_{ij} = \sum_{n\ge 0} n^{\underline{i}}(n+a)_j w_n. \tag{2.4}$$

Shifting $n = i+k$ and applying the ordinary Gauss sum gives

$$M_{ij} = \frac{\Gamma(a+i+j)\Gamma(c+i)\Gamma(d-i-j)}{\Gamma(b-a-j)\Gamma(b-c)}. \tag{2.5}$$

Indeed the hypergeometric numerator parameters are $a+i+j, c+i$, the denominator parameter is $b+i$, and the excess is $d-i-j$. Its real part is positive throughout the present index range.

Factor from row i the quantity

$$R_i = \frac{\Gamma(a+i)\Gamma(c+i)\Gamma(d-i)}{\Gamma(b-a)\Gamma(b-c)},$$

and from column j the factor $(1-b+a)_j$. The remaining entry is $(a+i)_j/(1-d+i)_j$. Lemma 2.2 applies with $x_i = a+i$ and $t = 1-b+c$. Since $x_k - x_i = k-i$, its Vandermonde is $\prod_{j=0}^{r-1} j!$. The factorial identities

$$(1-b+a)_j = (-1)^j\frac{\Gamma(b-a)}{\Gamma(b-a-j)}, \quad (1-b+c)_j = (-1)^j\frac{\Gamma(b-c)}{\Gamma(b-c-j)}$$

and

$$(1-d+i)_{r-1} = (-1)^{r-1}\frac{\Gamma(d-i)}{\Gamma(d-i-r+1)}$$

produce exactly (2.2); the total sign is one. Perform the calculation first away from the auxiliary determinant's poles, then extend by the identity theorem. The original sum and the final product are holomorphic on the asserted domain. □

Remark 2.3 (The one-variable boundary case). For $r = 1$, (2.2) is simply

$$\sum_{n\geq 0} w_n(a,c;b) = \frac{\Gamma(a)\Gamma(c)\Gamma(b-a-c)}{\Gamma(b-a)\Gamma(b-c)}.$$

The subsequent $r = 1$ resonance is the familiar zero-balanced Gauss case. The continuation of interest is the uniform mechanism for all r, not a claim of novelty for this boundary case.

3 First resonance and the correct finite part

Fix $r \geq 1$, initially $a, c > 0$, and set

$$b_0 = a + c + 2r - 2, \qquad b = b_0 + u. \tag{3.1}$$

Let $L(u)$ be any holomorphic function near zero with $L(0) = 1$. It is a normalization of *each* particle's weight. Define

$$\begin{aligned} A(u) &= L(u)^r Z_{r-1}(a,c;b_0+u), \\ E(u) &= \frac{\Gamma(r)\Gamma(a+r-1)\Gamma(c+r-1)\Gamma(1+u)^2}{\Gamma(r+u)\Gamma(a+r-1+u)\Gamma(c+r-1+u)}. \end{aligned} \tag{3.2}$$

Both are holomorphic near zero, and $E(0) = 1$. Although A uses the $(r-1)$-particle partition, its normalization is L^r, not L^{r-1}: the escaping largest index contributes a normalization factor as well.

Write the contribution of a prescribed largest index as

$$\begin{aligned} K_n(u) = L(u)^r w_n(a,c;b_0+u) \\ \times \sum_{0\leq m_1<\cdots<m_{r-1}<n} \Delta(\boldsymbol{m})^2 \prod_{i=1}^{r-1}(n-m_i)^2 w_{m_i}(a,c;b_0+u). \end{aligned} \tag{3.3}$$

For $r = 1$ the inner sum is one, including at $n = 0$; for $r > 1$ it vanishes if $n < r - 1$. Each K_n is holomorphic near zero.

Theorem 3.1 (Largest-index subtraction)**.** *For every fixed α with $\operatorname{Re}\alpha > 0$, the series*

$$D(u;\alpha) = \sum_{n\geq 0}\Big\{K_n(u) - A(u)(n+\alpha)^{-1-u}\Big\} \tag{3.4}$$

is locally normally convergent near $u = 0$, and

$$\boxed{D(u;\alpha) = A(u)\left\{\frac{E(u)}{u} - \zeta(1+u,\alpha)\right\}.} \tag{3.5}$$

The apparent pole on the right is removable. Every fixed derivative in u can be taken termwise. The assertions hold jointly locally holomorphically for $\operatorname{Re} a, \operatorname{Re} c, \operatorname{Re}\alpha > 0$.

Proof. Dividing the products in Theorem 2.1 at particle numbers r and $r-1$, with the same $b = b_0 + u$, gives

$$L(u)^r Z_r(a, c; b_0 + u) = \frac{A(u)E(u)}{u}. \tag{3.6}$$

For instance, the ratio of the factors common to the two products contains

$$\prod_{j=0}^{r-2} \frac{\Gamma(r-1+u-j)}{\Gamma(r+u-j)} = \frac{\Gamma(1+u)}{\Gamma(r+u)}.$$

The remaining factor has $\Gamma(u) = \Gamma(1+u)/u$. These two factors explain the square of $\Gamma(1+u)$ in E. Equation (3.6) initially holds for $\operatorname{Re} u > 0$ sufficiently small.

We next justify the subtraction independently of that product's meromorphic continuation. Work on a compact parameter set and a small closed disk $|u| \le \delta < 1/2$. The absolute $(r-1)$-particle sum has a uniformly finite first moment:

$$\sum_{m_1 < \cdots < m_{r-1}} (1 + m_1 + \cdots + m_{r-1}) |\Delta(\boldsymbol{m})|^2 \prod_i |w_{m_i}| < C. \tag{3.7}$$

Indeed, the largest exponent in the separate-index majorant is $-3 - \operatorname{Re} u$ before inserting a first moment, and at most $-2 + \delta < -1$ after insertion. This also proves the assertion uniformly for complex a, c in compact right-half-plane sets.

For $0 \le m_i < n$,

$$\left| 1 - \prod_i (1 - m_i/n)^2 \right| \le \frac{2}{n} \sum_i m_i.$$

The missing region $m_{r-1} \ge n$ is likewise bounded by the first moment divided by n. It follows that the inner sum in (3.3), divided by n^{2r-2}, is

$$Z_{r-1}(a, c; b_0 + u) + O(n^{-1})$$

uniformly on that compact set. The assertion is immediate for $r = 1$. The uniform gamma-ratio expansion gives

$$w_n(a, c; b_0 + u) = n^{-2r+1-u}\{1 + O(n^{-1})\}.$$

Consequently, for $n \ge 1$,

$$K_n(u) = A(u) n^{-1-u} + O(n^{-2+\delta}). \tag{3.8}$$

Replacing n by $n + \alpha$ in the leading term changes only the summable remainder, uniformly on compact α-sets. The finitely many initial terms are harmless. This proves local normal convergence of (3.4). Cauchy's integral formula on a slightly larger parameter disk proves normal convergence for every fixed u-derivative; there is no unproved exchange of an infinite series with differentiation at a pole.

For $\operatorname{Re} u > 0$, sum the leading counterterm by Hurwitz zeta and use (3.6). This yields (3.5) on a nonempty open set. Both sides extend holomorphically through zero, because $E(0) = 1$ and the Hurwitz residue is one. The identity theorem completes the proof, including joint holomorphy in the auxiliary parameters. □

Put $A_j = A^{(j)}(0)$ and

$$\chi = E'(0) = 2\psi(1) - \psi(r) - \psi(a + r - 1) - \psi(c + r - 1). \tag{3.9}$$

In particular,

$$A_0 = \prod_{j=0}^{r-2} \frac{j!\,\Gamma(a+j)\Gamma(c+j)\Gamma(r-j)}{\Gamma(c+2r-2-j)\Gamma(a+2r-2-j)}. \tag{3.10}$$

Corollary 3.2 (Hard-cutoff constant)**.** *For $a, c > 0$,*

$$\boxed{\lim_{N\to\infty}\left[\sum_{0\le n_1<\cdots<n_r\le N} \Delta(\boldsymbol{n})^2 \prod_i w_{n_i}(a,c;b_0) - A_0 \log N\right] = A_0\chi.} \tag{3.11}$$

The corresponding spectral finite part is

$$\mathrm{FP}_{u=0}\{L(u)^r Z_r(a,c;b_0+u)\} = A_1 + A_0\chi. \tag{3.12}$$

Thus the hard-cutoff constant is obtained from the spectral finite part by subtracting A_1.

Proof. At zero, Theorem 3.1 gives

$$D(0;\alpha) = A_0\{\chi - \gamma_0(\alpha)\} = A_0\{\chi + \psi(\alpha)\}.$$

The partial sum of the subtracted counterterm is $A_0\{\psi(N+1+\alpha) - \psi(\alpha)\}$. Since $\psi(N+1+\alpha) = \log N + o(1)$, the auxiliary α cancels and proves (3.11). Expanding $A(u)E(u)/u$ directly gives (3.12). □

Remark 3.3 (Why the normalization matters)*.* Changing L while preserving $L(0) = 1$ leaves the entire cutoff sum unchanged. It changes A_1 by $rL'(0)A_0$ relative to the choice $L \equiv 1$, and changes the spectral finite part by the same amount. Therefore identifying spectral and cutoff constants without specifying the amplitude derivative cannot be a normalization-invariant statement. This is a correction to that possible extrapolation, not an allegation that the inspected manuscript makes the identification.

4 All-order Stieltjes identities and Hurwitz primitives

4.1 The complete jet formula

Let $e_j = E^{(j)}(0)$, so $e_0 = 1$ and $e_1 = \chi$. All these coefficients are explicit finite polygamma polynomials. More precisely, put

$$\ell_j = 2\psi^{(j-1)}(1) - \psi^{(j-1)}(r) - \psi^{(j-1)}(a+r-1) - \psi^{(j-1)}(c+r-1). \tag{4.1}$$

Then $\log E(u) = \sum_{j\ge1} \ell_j u^j/j!$ and $e_j = \mathcal{B}_j(\ell_1,\dots,\ell_j)$. In particular,

$$e_2 = \chi^2 + \ell_2, \qquad e_3 = \chi^3 + 3\chi\ell_2 + \ell_3.$$

The derivatives A_j are obtained in exactly the same way from the finite Gamma product (3.2).

Theorem 4.1 (Every fixed Stieltjes order)**.** *For every integer $k \ge 0$, the following single-bracket series is ordinary and absolutely convergent:*

$$\sum_{n\ge0}\left[K_n^{(k)}(0) - \frac{1}{n+\alpha}\sum_{j=0}^{k}\binom{k}{j}A_j[-\log(n+\alpha)]^{k-j}\right]$$
$$= \sum_{j=0}^{k}\binom{k}{j}A_j\left\{\frac{e_{k-j+1}}{k-j+1} - (-1)^{k-j}\gamma_{k-j}(\alpha)\right\}. \tag{4.2}$$

The result is locally holomorphic for the parameters in Theorem 3.1. The constants A_j, e_j are finite Gamma/polygamma expressions; no infinite coefficient extraction is needed to compute a fixed row.

Proof. Differentiate the already normally convergent identity (3.5), rather than the divergent unsubtracted partition. The expansion at zero is

$$\frac{E(u)}{u} - \zeta(1+u,\alpha) = \sum_{h\geq 0} \left\{ \frac{e_{h+1}}{h+1} - (-1)^h \gamma_h(\alpha) \right\} \frac{u^h}{h!}.$$

Leibniz's rule gives the right side of (4.2). Differentiating $(n+\alpha)^{-1-u}$ gives the logarithmic polynomial on the left. The normal-convergence statement proved in Theorem 3.1 supplies absolute convergence of the resulting bracket and justifies every interchange. □

The first three values $D_k(\alpha) = \partial_u^k D(0;\alpha)$ are

$$\begin{aligned} D_0(\alpha) &= A_0\{\chi - \gamma_0(\alpha)\}, \\ D_1(\alpha) &= A_1\{\chi - \gamma_0(\alpha)\} + A_0\{e_2/2 + \gamma_1(\alpha)\}, \\ D_2(\alpha) &= A_2\{\chi - \gamma_0(\alpha)\} + 2A_1\{e_2/2 + \gamma_1(\alpha)\} + A_0\{e_3/3 - \gamma_2(\alpha)\}. \end{aligned} \tag{4.3}$$

The factor $1/(h+1)$ in these rows comes from dividing $E(u)-1$ by u before differentiation. Replacing it by one would give an incorrect higher-order table.

4.2 Derivatives in the subtraction point

The parameter α specifies the subtraction grid. It does not move the particles or change their original weights. Differentiating this auxiliary family nevertheless yields useful exact identities between the corresponding subtracted sums and the usual Hurwitz order ladder.

Proposition 4.2 (Hurwitz derivative and shift laws)**.** *For every integer $h \geq 1$,*

$$\partial_\alpha^h D(u;\alpha) = (-1)^{h+1} A(u)(1+u)_h \zeta(1+u+h,\alpha). \tag{4.4}$$

Moreover,

$$D(u;\alpha+1) - D(u;\alpha) = A(u)\alpha^{-1-u}. \tag{4.5}$$

Every fixed u-derivative of these equations is valid, including at $u=0$. In particular $\partial_\alpha^h D_0(\alpha) = A_0\psi^{(h)}(\alpha)$.

Proof. Use the classical identities $\partial_\alpha^h \zeta(s,\alpha) = (-1)^h (s)_h \zeta(s+h,\alpha)$ and $\zeta(s,\alpha+1) = \zeta(s,\alpha) - \alpha^{-s}$ [9]. Their substitution in (3.5) proves both formulas. Joint holomorphy, already established, permits the indicated mixed derivatives. The last formula also follows immediately from $D_0 = A_0(\chi + \psi(\alpha))$. □

4.3 Anchored antiderivatives and log-Gamma

An anchored primitive fixes the integration constant, which is especially important when differentiating in the spectral variable. Let $\operatorname{Re}\alpha_0, \operatorname{Re}\alpha > 0$, with the path in that half-plane.

Theorem 4.3 (Every anchored primitive of the spectral jets)**.** *The identity*

$$\int_{\alpha_0}^{\alpha} D(u;t)\,\mathrm{d}t = \frac{A(u)}{u}\left\{(\alpha-\alpha_0)E(u) + \zeta(u,\alpha) - \zeta(u,\alpha_0)\right\} \tag{4.6}$$

has a removable singularity at $u = 0$. For every $k \geq 0$,

$$\int_{\alpha_0}^{\alpha} D_k(t)\,\mathrm{d}t = \sum_{j=0}^{k}\binom{k}{j}\frac{A_j}{k-j+1}\big[(\alpha-\alpha_0)e_{k-j+1} + \zeta^{(k-j+1)}(0,\alpha) - \zeta^{(k-j+1)}(0,\alpha_0)\big]. \tag{4.7}$$

In particular,

$$\int_{\alpha_0}^{\alpha} D_0(t)\,\mathrm{d}t = A_0\left\{(\alpha-\alpha_0)\chi + \log\Gamma(\alpha) - \log\Gamma(\alpha_0)\right\}. \tag{4.8}$$

Proof. For $u \neq 0$, integrate the identity $\partial_t\zeta(u,t) = -u\zeta(1+u,t)$. This gives (4.6). The numerator vanishes at zero because $E(0) = 1$ and $\zeta(0,t) = \frac{1}{2} - t$. Its divided Taylor series has hth derivative at zero

$$\frac{(\alpha-\alpha_0)e_{h+1} + \zeta^{(h+1)}(0,\alpha) - \zeta^{(h+1)}(0,\alpha_0)}{h+1}.$$

Leibniz's rule proves (4.7). Finally, $\zeta'(0,t) = \log\Gamma(t) - \frac{1}{2}\log(2\pi)$ [9] gives (4.8). □

This theorem transports the classical Stieltjes primitive to a many-variable resonant harmonic sum. It does not claim a new formula for $\int \gamma_k(t)\,\mathrm{d}t$ in isolation, nor does it integrate an arbitrary many-variable weight with respect to a particle coordinate. The distinction identifies precisely which antiderivative has been established.

5 Integer and mixed harmonic specializations

5.1 A rational spectral partition at every particle number

Take $a = c = 1$, so $b_0 = 2r$, and choose

$$L(u) = \frac{\Gamma(2r+u)}{\Gamma(2r)}. \tag{5.1}$$

At zero the particle weight is rational:

$$w_n(1,1;2r) = \frac{1}{(n+1)_{2r-1}}.$$

Denote the normalized meromorphic partition by $F_r(u) = L(u)^r Z_r(1,1;2r+u)$.

Proposition 5.1 (Rationality of the normalized partition)**.** *For every $r \geq 1$,*

$$\boxed{F_r(u) = \frac{\prod_{j=0}^{r-1}(j!)^3}{[(2r-1)!]^r}\prod_{k=0}^{r-1}\frac{(u+r+k)_{r-k}}{(u+k)_r}.} \tag{5.2}$$

Its residue A_0 at zero is a positive rational number. The hard-cutoff constant is

$$A_0\{\gamma - 3H_{r-1}\}. \tag{5.3}$$

Proof. For $a = c = 1$, the jth product factor in (2.2) is

$$(j!)^3 \frac{\Gamma(r-1+u-j)}{\Gamma(2r-1+u-j)^2}.$$

Pair one factor $\Gamma(2r+u)$ from L^r with each of these, use Gamma recurrences, and replace j by $r-1-k$. This gives (5.2). Finally, $\chi = 2\psi(1) - 3\psi(r) = \gamma - 3H_{r-1}$, and Corollary 3.2 applies. □

The first partitions are

$$\begin{aligned}
F_1(u) &= \frac{u+1}{u}, \\
F_2(u) &= \frac{(u+3)^2}{36u(u+1)^2}, \\
F_3(u) &= \frac{(u+4)(u+5)^3}{216000u(u+1)^2(u+2)^3(u+3)}, \\
F_4(u) &= \frac{(u+6)^2(u+7)^4}{373403520000u(u+1)^2(u+2)^3(u+3)^4(u+4)^2}.
\end{aligned} \tag{5.4}$$

The corresponding residues are

$$A_0 = 1, \quad \frac{1}{4}, \quad \frac{1}{10368}, \quad \frac{1}{44789760000} \qquad (r = 1, 2, 3, 4). \tag{5.5}$$

Although F_r is rational, the individual normal-convergence amplitude $A(u)$ is generally not rational. Its polygamma derivatives are exactly what permit Stieltjes constants to appear in ordinary subtracted sums.

For calculation, the logarithmic derivatives are

$$(\log A)^{(k)}(0) = r\psi^{(k-1)}(2r) + \sum_{j=0}^{r-2} \left\{ \psi^{(k-1)}(r-j) - 2\psi^{(k-1)}(2r-1-j) \right\}, \tag{5.6}$$

$$\ell_k = (-1)^k (k-1)! \{3H_{r-1}^{(k)} - \zeta(k)\}, \qquad k \geq 2. \tag{5.7}$$

Together with $\ell_1 = \gamma - 3H_{r-1}$, these are finite exact coefficient recipes for every row of Theorem 4.1.

5.2 The derivatives as actual harmonic polynomials

For a configuration $\boldsymbol{n} = (n_1, \ldots, n_r)$ define

$$v_k(\boldsymbol{n}) = (-1)^k (k-1)! \sum_{i=1}^{r} \left\{ H_{n_i+2r-1}^{(k)} - H_{2r-1}^{(k)} \right\}. \tag{5.8}$$

Logarithmic differentiation of the normalized weight gives

$$K_n^{(k)}(0) = \sum_{0 \leq n_1 < \cdots < n_{r-1} < n_r = n} \frac{\Delta(\boldsymbol{n})^2 \, \mathcal{B}_k(v_1(\boldsymbol{n}), \ldots, v_k(\boldsymbol{n}))}{\prod_{i=1}^{r} (n_i+1)_{2r-1}}. \tag{5.9}$$

This formula has no spectral derivatives left unevaluated. To see it, for a single particle differentiate $\log \Gamma(2r+u) - \log \Gamma(n+2r+u)$. The polygamma shift formula gives the corresponding summand in

(5.8); the Bell identity (1.3) then differentiates the product. Substitution in (4.2) proves an explicit all-order family of harmonic–logarithm–Stieltjes identities.

For a concrete first-order member, put

$$W_{m,n} = \frac{(n-m)^2}{(m+1)_3(n+1)_3}, \qquad 0 \le m < n. \tag{5.10}$$

At $r = 2$, $A_0 = 1/4$, $A_1 = (5/3 - \gamma)/4$, and $e_2 = (\gamma - 3)^2 + 3 - \zeta(2)$. Thus

$$\sum_{n\ge 0}\Bigg[\sum_{m=0}^{n-1} W_{m,n}(2H_3 - H_{m+3} - H_{n+3}) - \frac{5/3 - \gamma - \log(n+1)}{4(n+1)}\Bigg] = \frac{1}{4} + \frac{\gamma^2 - \zeta(2)}{8} + \frac{\gamma_1}{4}. \tag{5.11}$$

Here $\gamma_1 = \gamma_1(1)$, and the empty inner sum at $n = 0$ is zero. The zero-order companion is

$$\sum_{n\ge 0}\left\{\sum_{m<n} W_{m,n} - \frac{1}{4(n+1)}\right\} = -\frac{3}{4}. \tag{5.12}$$

These are ordinary convergent bracketed series, not a declaration of finite-part integration in disguise.

The same example makes the normalization distinction numerical and exact:

$$\operatorname{FP}_{u=0} F_2(u) = -\frac{1}{3}, \qquad \lim_{N\to\infty}\left\{\sum_{m<n\le N} W_{m,n} - \frac{1}{4}\log N\right\} = \frac{\gamma - 3}{4}. \tag{5.13}$$

Their difference is $A_1 = (5/3 - \gamma)/4$, as required. With $L = 1$ the spectral finite part instead equals $(2\gamma - 5)/4$; the hard cutoff is unchanged.

5.3 Ordinary all-order mixed harmonic sums

A convergent, rather than resonant, specialization produces another useful family. For an integer $B > 2r$, define

$$\mathcal{H}_{r,B}(p,q) = \frac{Z_r(1+p, 1+q; B)}{[\Gamma(1+p)\Gamma(1+q)]^r}. \tag{5.14}$$

Near $p = q = 0$, it is both a normally convergent sum and the explicit Gamma product

$$\mathcal{H}_{r,B}(p,q) = \prod_{j=0}^{r-1} \frac{j!\,\Gamma(1+p+j)\Gamma(1+q+j)\Gamma(B-r-1-j-p-q)}{\Gamma(1+p)\Gamma(1+q)\Gamma(B-1-j-p)\Gamma(B-1-j-q)}. \tag{5.15}$$

The convergence region includes a neighborhood with $\operatorname{Re}(p+q) < B - 2r$.

Theorem 5.2 (Many-variable mixed harmonic identities)**.** *For a configuration $\boldsymbol{n}$, let*

$$U_j(\boldsymbol{n}) = (-1)^{j-1}(j-1)!\sum_{i=1}^{r} H_{n_i}^{(j)}, \qquad W_B(\boldsymbol{n}) = \frac{\Delta(\boldsymbol{n})^2}{\prod_i (n_i+1)_{B-1}}.$$

For all $k, l \ge 0$,

$$\sum_{n_1<\cdots<n_r} W_B(\boldsymbol{n})\mathcal{B}_k(U_1,\dots,U_k)\mathcal{B}_l(U_1,\dots,U_l) = \partial_p^k \partial_q^l \mathcal{H}_{r,B}(0,0). \tag{5.16}$$

Every pure row ($k = 0$ or $l = 0$) is rational. Every mixed row lies in $\mathbb{Q}[\zeta(2), \dots, \zeta(k+l)]$; for zero or first total order this means simply $\mathbb{Q}$.

Proof. Relative to its value at $p = q = 0$, the summand is multiplied by

$$\prod_{i=1}^{r} \prod_{j=1}^{n_i} (1 + p/j)(1 + q/j).$$

Its logarithmic derivatives in either variable are the U_j. The Bell formula and local normal convergence prove (5.16). The pure generating function is itself rational:

$$\mathcal{H}_{r,B}(p, 0) = \prod_{j=0}^{r-1} \frac{(j!)^2 (1+p)_j}{\Gamma(B-1-j)(B-r-1-j-p)_r}. \tag{5.17}$$

This follows by Gamma recurrences in (5.15). It proves rationality of every pure derivative, not just the displayed low orders. For derivatives involving both p and q, only $\log \Gamma(B-r-1-j-p-q)$ contributes to a mixed derivative of the logarithm. Its derivatives of total order $h \geq 2$ are rational combinations of $\zeta(h)$ and finite generalized harmonic numbers. Pure logarithmic derivatives are rational by (5.17). Exponentiating with finite multivariate Bell coefficients proves the asserted zeta algebra. □

For $r = 2, B = 6$, the product simplifies to

$$\mathcal{H}_{2,6}(p, q) = \frac{(1+p)(1+q)\Gamma(3-p-q)\Gamma(2-p-q)}{\Gamma(5-p)\Gamma(5-q)\Gamma(4-p)\Gamma(4-q)}. \tag{5.18}$$

Set $V_{m,n} = (n-m)^2/[(m+1)_5 (n+1)_5]$ and sum below over $0 \leq m < n$. Direct differentiation yields

$$\begin{aligned}
\sum V_{m,n} &= \frac{1}{10368}, \\
\sum V_{m,n}(H_m + H_n) &= \frac{29}{12 \cdot 10368}, \\
\sum V_{m,n}\{(H_m + H_n)^2 - H_m^{(2)} - H_n^{(2)}\} &= \frac{43}{8 \cdot 10368}, \\
\sum V_{m,n}(H_m + H_n)^2 &= \frac{517/144 + 2\zeta(2)}{10368}, \\
\sum V_{m,n}\{H_m^{(2)} + H_n^{(2)}\} &= \frac{2\zeta(2) - 257/144}{10368}.
\end{aligned} \tag{5.19}$$

For example, at zero the first logarithmic p-derivative is $29/12$, the second pure logarithmic derivative is $-67/144$, and the mixed logarithmic derivative is $2\zeta(2) - 9/4$. These three numbers prove all four nonconstant rows. They illustrate why a pure Bell combination can be rational while the square of the ordinary harmonic sum is not.

6 Rank-one harmonic dependence and depth-two closure

6.1 A theorem beyond consecutive factorial weights

The next result is algebraic and does not require the Gamma product. Let $w(x) \in \mathbb{Q}(x)$ have only simple poles, all at negative integers, and suppose

$$w(n) = \lambda n^{-(2r-1)} + O(n^{-2r}), \qquad \lambda \in \mathbb{Q} \setminus \{0\}. \tag{6.1}$$

There is no positivity assumption. The pole locations need not be consecutive. Define

$$S_{w,N} = \sum_{0 \le n_1 < \cdots < n_r \le N} \Delta(\boldsymbol{n})^2 \prod_i w(n_i), \qquad S_{w,-1} = 0. \tag{6.2}$$

Theorem 6.1 (Rank-one harmonic dependence)**.** *There exist rational functions $P_w(N), Q_w(N) \in \mathbb{Q}(N)$ such that for every integer $N \ge 0$,*

$$\boxed{S_{w,N} = P_w(N) H_{N+1} + Q_w(N).} \tag{6.3}$$

Both rational functions are bounded at infinity. At each finite pole, P_w has order at most $r-1$ and Q_w has order at most r. The possible poles are negative integers at most -2. For $r = 1$, $P_w = \lambda$.

Proof. For $0 \le k \le 2r-2$, partial fractions give

$$n^k w(n) = \sum_{d \ge 1} \frac{c_{k,d}}{n+d}, \qquad \sum_d c_{k,d} = \lambda \mathbf{1}_{k=2r-2}. \tag{6.4}$$

Only finitely many coefficients are nonzero. There is no polynomial part because $n^k w(n)$ is proper. The sum of residues is the coefficient of n^{-1} at infinity, proving the second equality.

Thus the finite moment $M_k(N) = \sum_{n=0}^N n^k w(n)$ equals

$$\begin{aligned} M_k(N) &= R_k(N) + \lambda \mathbf{1}_{k=2r-2} H_{N+1}, \\ R_k(N) &= \sum_d c_{k,d} \left\{ \sum_{\ell=2}^{d} \frac{1}{N+\ell} - H_{d-1} \right\}. \end{aligned} \tag{6.5}$$

Every R_k is a bounded rational function with simple poles only at the indicated negative integers. Finite Cauchy–Binet gives

$$S_{w,N} = \det_{0 \le i,j < r} M_{i+j}(N). \tag{6.6}$$

Among these entries, $i + j = 2r - 2$ occurs only at $i = j = r - 1$. Therefore

$$(M_{i+j})_{i,j=0}^{r-1} = (R_{i+j})_{i,j=0}^{r-1} + \lambda H_{N+1} \boldsymbol{e}_{r-1} \boldsymbol{e}_{r-1}^{\mathsf{T}}. \tag{6.7}$$

The determinant is affine in the harmonic number, with

$$P_w = \lambda \det_{0 \le i,j < r-1} R_{i+j}, \qquad Q_w = \det_{0 \le i,j < r} R_{i+j}. \tag{6.8}$$

The empty-determinant convention covers $r = 1$. Products of $r-1$ or r simple-pole entries prove the pole-order bounds, and boundedness of every entry proves boundedness at infinity. This argument proves (6.3) at every cutoff, including $N < r-1$ where the determinant and configuration sum both vanish. □

Remark 6.2 (The source of the depth collapse). A generic determinant whose entries are affine in H_{N+1} has harmonic degree as large as r. That estimate is much too weak here. The decay condition (6.1) forces the residue sum to vanish in every moment except the highest one. Consequently the harmonic matrix has rank one *and a single nonzero entry.* This is the mechanism, not cancellation observed in a few low-rank examples.

6.2 An explicit partial-fraction transform

For later use put

$$T_{p,j}(z)=\sum_{m\ge 1}\frac{z^m}{(m+j)^p},\qquad U_{p,j}(z)=\sum_{m\ge 1}\frac{z^m H_m}{(m+j)^p},\qquad p\ge 1,\ j\ge 0.$$

Shifting the summation index gives

$$T_{p,j}(z)=z^{-j}\left\{\mathrm{Li}_p(z)-\sum_{k=1}^{j}\frac{z^k}{k^p}\right\}. \tag{6.9}$$

For the harmonic sums the precise shifted formula is

$$U_{p,0}(z)=\mathrm{Li}_{p,1}(z,1)+\mathrm{Li}_{p+1}(z), \tag{6.10}$$

$$U_{p,j}(z)=z^{-j}\left\{\mathrm{Li}_{p,1}(z,1)-\sum_{k=1}^{j}\frac{z^k H_{k-1}}{k^p}-\sum_{\ell=1}^{j-1}\sum_{k=j+1}^{\infty}\frac{z^k}{k^p(k-\ell)}\right\},\quad j\ge 1. \tag{6.11}$$

For (6.11), write $k=m+j$ and use $H_m=H_{k-1}-\sum_{\ell=1}^{j-1}(k-\ell)^{-1}$. The finite correction is H_{k-1}, not H_k. Each last rational tail has a partial-fraction decomposition in k and $k-\ell$, hence involves only ordinary polylogarithms and finite sums. Its starting index $j+1>\ell$ avoids the apparent pole. Equations (6.9)–(6.11) are explicit terminating reduction rules, not formal existence statements.

Theorem 6.3 (All-particle depth-two closure)**.** *Under the assumptions of Theorem 6.1, let* $K_{w,N}=S_{w,N}-S_{w,N-1}$ *and*

$$\mathcal{G}_w(z)=\sum_{N\ge 0}K_{w,N}z^{N+1},\qquad |z|<1. \tag{6.12}$$

Then $\mathcal{G}_w$ *belongs to the* $\mathbb{Q}(z)$*-linear span of*

$$1,\quad \mathrm{Li}_1(z),\dots,\mathrm{Li}_r(z),\quad \mathrm{Li}_{1,1}(z,1),\dots,\mathrm{Li}_{r-1,1}(z,1). \tag{6.13}$$

For $r=1$ *the depth-two part of this list is empty. Thus the depth upper bound is two, independently of* r*, and the polylogarithmic weight upper bound is* r*.*

Proof. Set $m=N+1$. By (6.3),

$$\begin{aligned}K_{w,N}=&\{P_w(N)-P_w(N-1)\}H_{N+1}\\&+Q_w(N)-Q_w(N-1)+\frac{P_w(N-1)}{N+1}.\end{aligned} \tag{6.14}$$

The harmonic coefficient is a proper rational function of m, with poles only at nonpositive integers and pole orders at most $r-1$. It is zero for $r=1$. The nonharmonic coefficient is also proper, with pole orders at most r: division by m can increase the order of $P_w(N-1)$ at zero only from at most $r-1$ to at most r. All other poles already have the stated bound. The expressions at $N=-1$ used in (6.14) are well defined; (6.5) gives the empty moments there.

Decompose both coefficients into terms $(m+j)^{-p}$ with $j\ge 0$. For the nonharmonic part apply (6.9), which uses indices at most r. For the harmonic part apply (6.10) and (6.11). The resulting depth-two index is at most $r-1$; any additional ordinary polylogarithm has index at most r. This proves (6.13). All the series converge absolutely in the open disk, so the shifts and finite partial fractions are legitimate. □

The equivalent identity

$$\mathcal{G}_w(z) = (1-z)\sum_{N\geq 0} S_{w,N} z^{N+1} \tag{6.15}$$

connects the cutoff and Abel descriptions. It is useful for checking constants but is not needed to guess the depth bound.

6.3 Derivatives, Euler primitives, and endpoint zeta reduction

Let $\Theta = z\,d/dz$ and normalize Θ^{-1} by integration from zero with the kernel dt/t. Then

$$\Theta^{-q}\mathcal{G}_w(z) = \sum_{N\geq 0} \frac{K_{w,N}}{(N+1)^q} z^{N+1}, \qquad q \geq 0. \tag{6.16}$$

This also equals, for $q \geq 1$,

$$\frac{1}{(q-1)!}\int_0^z \log^{q-1}(z/t)\mathcal{G}_w(t)\frac{\mathrm{d}t}{t}, \tag{6.17}$$

initially along $0 < t < z < 1$, then as the analytic primitive of the power series.

Corollary 6.4 (Primitive closure and ordinary endpoint evaluations)**.** *For every $q \geq 0$, the function (6.16) has polylogarithmic depth at most two and weight at most $r+q$. For $q \geq 1$ its endpoint*

$$I_{w,q} = \sum_{N\geq 0} \frac{K_{w,N}}{(N+1)^q} \tag{6.18}$$

is an ordinary absolutely convergent sum and lies in the rational linear span of

$$1, \qquad \zeta(k), \qquad \zeta(a)\zeta(b), \quad 2 \leq k \leq r+q, \quad a,b \geq 2, \quad a+b \leq r+q.$$

No product of more than two zeta factors is needed for this endpoint family. Ordinary derivatives and anchored ordinary antiderivatives also retain depth at most two, although their rational prefactors and exponent shifts change.

Proof. Divide the two proper rational coefficients in (6.14) by m^q. Their maximum pole orders become at most $r-1+q$ in the harmonic part and $r+q$ in the other part. The preceding proof therefore gives the asserted weight and depth bounds.

Bounded rational functions have expansions at infinity, so (6.14) also gives $K_{w,N} = P_w(\infty)/(N+1) + O(\log(N+2)/(N+1)^2)$. This proves absolute convergence of (6.18). The configuration sum itself is absolutely convergent after inserting the maximum weight $(n_r+1)^{-q}$, by the same absolute largest-index estimate used in Section 3.

The reduction rules give Laurent-polynomial, hence locally analytic near $z=1$, coefficients of the polylogarithms. Only $\mathrm{Li}_1(z)$ and $\mathrm{Li}_{1,1}(z,1) = \frac{1}{2}\log^2(1-z)$ diverge there. Convergence forces their coefficients to vanish at one: a nonzero quadratic logarithm cannot cancel a linear logarithm or a bounded term, and a nonzero linear logarithm cannot cancel bounded terms. Those vanishing-coefficient terms tend to zero. The other endpoint values are $\zeta(p)$ and $\zeta(p,1)$. Lemma 6.5 below reduces the latter to the stated span.

For the last assertion, power-series differentiation multiplies coefficients by affine factors in the index and shifts the exponent; ordinary integration divides by such factors and shifts it oppositely. A finite shift of H_m changes it by a rational function. Polynomial parts in the index, when they

arise, sum to rational functions or rational multiples of $\mathrm{Li}_1(z)$. Partial fractions of the remaining proper parts again use (6.9)–(6.11), with finite initial terms removed where needed. None of these operations requires a third nested index. □

Lemma 6.5 (Euler's height-one reduction)**.** *For every integer $p \geq 2$,*

$$\zeta(p,1) = \frac{p}{2}\zeta(p+1) - \frac{1}{2}\sum_{j=1}^{p-2}\zeta(p-j)\zeta(j+1). \tag{6.19}$$

Proof. We include a beta-integral proof of this classical identity so the endpoint theorem has no unproved multiple-zeta reduction step. For $|x| < 1$, expand Euler's beta integral [7]:

$$\begin{aligned} B(1-x,y) - B(1,y) &= \int_0^1 (t^{-x}-1)(1-t)^{y-1}\mathrm{d}t \\ &= -\gamma - \psi(1-x) \\ &\quad + \frac{y}{2}\{[\gamma+\psi(1-x)]^2 + \zeta(2) - \psi'(1-x)\} + O(y^2). \end{aligned}$$

The y-derivative at zero can be taken under the integral: the factor $t^{-x}-1$ cancels its endpoint singularity at $t=1$. Using $\log(1-t)/(1-t) = -\sum_{n\geq 1} H_n t^n$ gives this derivative as

$$-\sum_{n\geq 1} H_n \left\{\frac{1}{n+1-x} - \frac{1}{n+1}\right\} = -\sum_{p\geq 2}\zeta(p,1)x^{p-1}.$$

The interchanges hold first for real $0 < x < 1$ by absolute convergence, and then throughout the disk by holomorphy. Put $f(x) = -\psi(1-x) - \gamma = \sum_{k\geq 2}\zeta(k)x^{k-1}$. Comparison with the beta expansion yields

$$\sum_{p\geq 2}\zeta(p,1)x^{p-1} = \tfrac{1}{2}\{f'(x) - \zeta(2) - f(x)^2\}.$$

Equating coefficients of x^{p-1} proves (6.19). □

6.4 A common translate and cyclotomic coordinates

The rank-one mechanism survives translation of the entire simple-pole grid. Define

$$H_n(a) = \sum_{j=0}^{n-1}\frac{1}{j+a}, \qquad H_0(a) = 0, \qquad a > 0.$$

Suppose all poles of $w(n)$ are simple and lie at $-a-d$ with integers $d \geq 0$, and the critical decay (6.1) holds. Use the monic moment basis $(n+a)^k$. For each pole,

$$\sum_{n=0}^{N}\frac{1}{n+a+d} = H_{N+1}(a) + \sum_{\ell=1}^{d}\frac{1}{N+a+\ell} - \sum_{\ell=0}^{d-1}\frac{1}{a+\ell}.$$

The sum of residues still vanishes except in the top moment. The proof of Theorem 6.1 therefore gives

$$S_{w,N} = P_w(N,a)H_{N+1}(a) + Q_w(N,a), \tag{6.20}$$

with the same pole-order bounds. For a rational weight depending rationally on a, the coefficients are rational in N, a.

For rational $a = p/q \in (0,1]$ in lowest terms, put $\omega = e^{2\pi i/q}$. The exact root-of-unity filters are

$$\sum_{n\geq 0} \frac{z^{n+a}}{(n+a)^s} = q^{s-1} \sum_{j=0}^{q-1} \omega^{-pj} \operatorname{Li}_s(\omega^j z^{1/q}), \tag{6.21}$$

$$\sum_{n\geq 0} \frac{z^{n+a} H_n(a)}{(n+a)^s} = q^{s-1} \sum_{j,k=0}^{q-1} \omega^{-p(j+k)} \operatorname{Li}_{s,1}(\omega^j z^{1/q}, \omega^k). \tag{6.22}$$

These formulas hold for integers $s \geq 1$ and $0 < z < 1$, and on any common continuation with a fixed qth-root branch. To prove them, write the top index as $qn + p$ and the inner index as $qm + p$. The powers of q from the denominators are q^s and q^{s+1}; the one and two residue filters divide by q and q^2, respectively. The strict inner inequality is preserved.

Finite index shifts and partial fractions, exactly as in (6.11), now prove depth at most two in level-q colored polylogarithms for the generating function with exponent z^{N+a}. The same conclusion holds for primitives that divide its coefficients by $(N+a)^h$. No ordinary-zeta-only endpoint reduction is asserted for general level q.

The shifted canonical weight $w(n) = 1/(n+a)_{2r-1}$ is also covered by the Gamma product: take the parameters there to be $(a, c) = (a, 1)$ and $b_0 = a + 2r - 1$. For $r = 2$ its residue and cutoff constant are

$$A_0 = \frac{1}{2a(a+1)}, \qquad C_a = \frac{-2 - \psi(a+1)}{2a(a+1)}. \tag{6.23}$$

For example, at $a = 1/2$,

$$\lim_{N\to\infty} \left[\sum_{0\leq m<n\leq N} \frac{(n-m)^2}{(m+\frac{1}{2})_3 (n+\frac{1}{2})_3} - \frac{2}{3}\log N \right] = \frac{2}{3}\{\gamma + 2\log 2 - 4\}. \tag{6.24}$$

This gives a direct digamma/cyclotomic bridge without changing the finite-cutoff depth mechanism.

7 Explicit polylogarithms and ordinary multiple sums

In this section use the canonical critical weight $w_r(n) = 1/(n+1)_{2r-1}$. Abbreviate $S_{r,N} = S_{w_r,N}$, $K_{r,N} = S_{r,N} - S_{r,N-1}$, and

$$G_r(z) = \sum_{N\geq 0} K_{r,N} z^{N+1}, \qquad I_{r,q} = \sum_{0\leq n_1<\cdots<n_r} \frac{\Delta(\boldsymbol{n})^2}{(n_r+1)^q \prod_i (n_i+1)_{2r-1}}, \quad q \geq 1. \tag{7.1}$$

Thus $I_{r,q} = \Theta^{-q} G_r(1)$, an ordinary convergent endpoint. The one-variable case is $G_1 = \operatorname{Li}_1$ and $I_{1,q} = \zeta(q+1)$.

7.1 Two variables: complete cutoff and generating function

The three moments needed for $r = 2$ are

$$\begin{aligned} M_0(N) &= \frac{(N+1)(N+4)}{4(N+2)(N+3)}, \\ M_1(N) &= \frac{N(N+1)}{4(N+2)(N+3)}, \\ M_2(N) &= H_{N+1} - \frac{11}{4} + \frac{1}{2(N+2)} + \frac{9}{2(N+3)}. \end{aligned} \tag{7.2}$$

They follow directly from the residues of $n^k/(n+1)_3$. Taking $M_0M_2 - M_1^2$ gives the finite identity

$$\boxed{S_{2,N} = \frac{(N+1)(N+4)}{4(N+2)(N+3)}H_{N+1} - \frac{(N+1)^2(3N+8)}{4(N+2)^2(N+3)}.} \tag{7.3}$$

This proves the cutoff formula without any asymptotic summation. It also independently recovers the residue $1/4$ and constant $(\gamma - 3)/4$.

Applying the exact transforms in Section 6 yields

$$\begin{aligned} G_2(z) = {} & \frac{(1-z)^2}{4z^2}\log^2(1-z) + \frac{z-1}{2z}\operatorname{Li}_2(z) \\ & - \frac{6-5z}{4z^2}\log(1-z) + \frac{3}{4} - \frac{3}{2z}. \end{aligned} \tag{7.4}$$

The expression is analytic at zero and starts $z^2/144$, corresponding to the unique first configuration $(0,1)$. Every negative-power coefficient cancels exactly. At $z = 1/2$, the classical dilogarithm reflection formula [10] gives

$$\sum_{0\le m<n} \frac{(n-m)^2\, 2^{-(n+1)}}{(m+1)_3(n+1)_3} = \frac{1}{2}\log^2 2 + \frac{7}{2}\log 2 - \frac{\pi^2}{24} - \frac{9}{4}. \tag{7.5}$$

At the other endpoint, (7.4) gives

$$G_2(z) = -\frac{1}{4}\log(1-z) - \frac{3}{4} + o(1), \qquad z \uparrow 1. \tag{7.6}$$

The Abel constant is $-3/4$, not the hard-cutoff constant $(\gamma - 3)/4$. This is consistent with $\sum_{N\ge 0} H_{N+1}z^{N+1} = -\log(1-z)/(1-z)$.

Theorem 7.1 (An ordinary double-sum evaluation)**.**

$$\boxed{\sum_{0\le m<n} \frac{(n-m)^2}{(m+1)_3(n+1)_3(n+1)} = \zeta(3) - \frac{\pi^2}{24} - \frac{3}{4}.} \tag{7.7}$$

Proof. We give a telescoping proof in addition to the polylogarithmic primitive method. Put $v = N + 1$. Differencing (7.3) gives

$$K_{2,v-1} = \frac{4(v+1)H_v + v^3 - 3v^2 - 6v}{4v(v+1)^2(v+2)}.$$

Partial fractions of $K_{2,v-1}/v$ give a harmonic coefficient

$$-\frac{3}{4v} + \frac{1}{v+1} - \frac{1}{4(v+2)} + \frac{1}{2v^2}$$

and a nonharmonic coefficient

$$-\frac{3}{4v} + \frac{5}{4(v+1)} - \frac{1}{2(v+2)} + \frac{1}{2(v+1)^2}.$$

Group the first three terms of each coefficient into telescoping differences before summing. Absolute convergence then permits

$$\sum_{v\ge 1} \frac{H_v}{v(v+1)} = \zeta(2), \qquad \sum_{v\ge 1} \frac{H_v}{(v+1)(v+2)} = 1, \qquad \sum_{v\ge 1} \frac{H_v}{v^2} = 2\zeta(3).$$

The first two identities follow by summing over the inner index in H_v and telescoping; the third is $\zeta(2,1)+\zeta(3)=2\zeta(3)$ by Lemma 6.5. The harmonic part therefore sums to $-3\zeta(2)/4+1/4+\zeta(3)$. The nonharmonic part sums to $\zeta(2)/2-1$. Their total is $\zeta(3)-\zeta(2)/4-3/4$, proving the theorem. □

Two further primitives give

$$I_{2,2}=\frac{7}{4}-\frac{\pi^2}{16}-\frac{3}{2}\zeta(3)+\frac{\pi^4}{144}, \tag{7.8}$$

$$I_{2,3}=-\frac{5}{2}+\frac{17\pi^2}{96}+\zeta(3)-\frac{\pi^4}{96}-\frac{\pi^2\zeta(3)}{12}+\frac{3}{2}\zeta(5). \tag{7.9}$$

These follow by dividing the same rational-plus-harmonic mass by v^2 and v^3, applying (6.9)–(6.11), and then (6.19). The delivered exact formula file contains the full primitive reductions, not just their endpoints.

7.2 Three variables: a depth-two formula of weight three

For $r=3$, the determinant gives

$$P_3(N)=\frac{N(N+1)^2(N^2+17N+54)}{10368(N+3)^2(N+4)^2(N+5)} \tag{7.10}$$

and

$$Q_3(N)=-\frac{N(N+1)^3(9N^3+125N^2+514N+648)}{20736(N+2)(N+3)^3(N+4)^2(N+5)}, \tag{7.11}$$

with $S_{3,N}=P_3(N)H_{N+1}+Q_3(N)$. Hence its hard-cutoff constant is $(\gamma-9/2)/10368$, again agreeing with the independent Gamma resonance calculation.

For compact notation let $L_j=\mathrm{Li}_j(z)$ and $D_j=\mathrm{Li}_{j,1}(z,1)$. The generating function is

$$\begin{aligned}G_3(z)=&-\frac{5(z-1)^2(7z-4)}{1728z^4}D_1+\frac{(z-1)(z+1)}{144z^3}D_2\\&-\frac{117z^3+95z^2-513z+300}{10368z^4}L_1\\&-\frac{5(z-1)(7z-18)}{1728z^3}L_2+\frac{z-1}{144z^2}L_3-\frac{281z^2-78z-200}{6912z^3}.\end{aligned} \tag{7.12}$$

Its first nonzero coefficient is $z^3/54432000$, the configuration $(0,1,2)$. Thus three ordered summation variables have not produced depth three. The theorem explains this before any coefficient calculation is performed.

The first three ordinary primitive endpoints are

$$I_{3,1}=\frac{1903}{124416}-\frac{35\pi^2}{62208}-\frac{7}{864}\zeta(3), \tag{7.13}$$

$$I_{3,2}=-\frac{6775}{373248}+\frac{67\pi^2}{186624}+\frac{7}{576}\zeta(3), \tag{7.14}$$

$$I_{3,3}=\frac{100297}{4478976}-\frac{403\pi^2}{1492992}-\frac{131}{7776}\zeta(3)+\frac{\pi^4}{186624}. \tag{7.15}$$

In particular, (7.13) is the explicit triple sum

$$\sum_{0\le i<j<k}\frac{(j-i)^2(k-i)^2(k-j)^2}{(i+1)_5(j+1)_5(k+1)_5(k+1)}.$$

The displayed value is a reduction of this entire sum, not a bound or an asymptotic approximation.

7.3 Four variables and the uniform algorithm

A first four-variable endpoint is

$$I_{4,1} = \frac{630863}{268738560000} + \frac{14717\pi^2}{67184640000} - \frac{487}{124416000}\zeta(3) + \frac{\pi^4}{497664000}. \tag{7.16}$$

The definition (7.1) specifies its six squared difference factors and four degree-seven factorial denominators. The numbers in these expressions can cancel substantially; direct low-precision evaluation is therefore a poor way to assess them. The exact rational reduction precedes the high-precision diagnostic.

For any requested r and primitive order q, there is a terminating algorithm: compute the $2r-1$ finite moments by residues, take the two determinants (6.8), difference the cutoff, divide by $(N+1)^q$, and apply the finite partial-fraction transform. It outputs both a functional polylogarithmic identity and, when $q \geq 1$, its ordinary zeta endpoint. The implementation in `code/reductions.py` uses exact rational arithmetic. It is not restricted in its mathematics to the ranks and primitive orders printed here, although symbolic expression growth limits convenient computation at large rank.

The fact that some displayed endpoints omit the highest permitted weight is an additional cancellation in those cases. We assert the proved upper bound $r+q$, not a universal sharp-weight formula and not independence of the surviving zeta coordinates.

8 Source audit, corrections, and integration status

8.1 What was actually inspected

The mathematical source anchor for the selected chapter reads is the `ProveIt` commit

`87c7db629552afbc2e803001526debe3430a31b0`.

The inspected material comprises the canonical manuscript README, the mixed-Dougall chapter (including its full ending), the initial 240 lines of the cyclotomic-quotient chapter, the incoming-directory inventory, and the first 180 lines of the incoming intake procedure. Targeted repository searches supplied additional context about existing Gauss, Stieltjes, tail, and Selberg-related work. Search snippets were not treated as complete proofs of the unseen material.

The repository changed during retrieval: later search results carried a different commit. This report therefore records a source anchor, not an assertion that the whole repository was frozen throughout the session. Six ZIP archives were visible in the incoming inventory at the time it was read. Their binary contents were not unpacked or analytically audited in this work. File names, including date strings, do not establish their mathematical content. No statement here should be read as a certification of those archives or of the entire four-volume manuscript.

No substantive false theorem was found in the fully inspected mixed-Dougall proof. Its all-order mixed derivatives and tail identities remain prior repository material, not contributions claimed by this report. The already corrected Choi normalization is likewise not a newly discovered correction here [1, 2].

8.2 Specific correction and normalization safeguards

The principal correction supplied by the present analysis is the *transfer rule* between two possible definitions of a finite constant. For this family the correct formula is

$$\text{hard-cutoff constant} = \text{spectral finite part} - A'(0).$$

Equation (5.13) is an explicit counterexample to omitting the last term. The rule corrects that possible extrapolation; it is not attributed to any particular source statement that was not observed.

Two further safeguards are necessary when implementing the formulas. The leading amplitude uses $L(u)^r$, even though its partition has $r-1$ particles. In the shifted harmonic transform (6.11), the finite correction is H_{k-1}. Both choices follow from direct derivations, and the delivered coefficient checks are sensitive to the shift convention. The ordered-configuration normalization also precludes an extra factor of $r!$.

The finite cutoff, the Abel constant, the spectral finite part, and the value of an ordinary subtracted bracket are four distinct normalizations. For the canonical two-particle example they are, respectively,

$$\frac{\gamma-3}{4}, \qquad -\frac{3}{4}, \qquad -\frac{1}{3} \text{ for the normalization (5.1)}, \qquad -\frac{3}{4} \text{ at subtraction point } \alpha = 1.$$

The numerical equality of the second and fourth constants in this example does not identify all four operations.

8.3 Claim ledger

Result	Status in this report	Scope qualification
Gauss–Vandermonde product	Complete determinant proof	Classical inputs; no priority claim
Largest-index finite part	Complete analytic proof	First logarithmic resonance only
All Stieltjes orders	Complete normal-convergence proof	Fixed finite derivatives; full brackets
Harmonic mixed generator	Complete Gamma-product proof	Integer $B > 2r$, local parameter germ
Rank-one harmonic cutoff	Complete algebraic proof	Simple negative-integer poles and critical decay
Depth-two closure	Complete terminating reduction	Upper bound, not minimal depth
Zeta endpoint evaluations	Complete reduction with Euler lemma	Ordinary positive primitive orders
Common translated grid	Complete residue/filter proof	Common grid, not arbitrary unrelated shifts
S_6, S_8	Not resolved	Prior conjectural status unchanged
Proof-assistant status	Not formalized	No Lean or other kernel certificate

The source's distinction between formal relation systems and actual period independence is preserved [3]. A finite rank calculation or a failed integer-relation search would not prove that an endpoint lies outside a smaller period space. No such nonreduction assertion is used here.

8.4 Proposed integration

A suitable standalone destination is

`Analysis/Polylogarithms/docs/reports/determinantal-harmonic-resonance/`.

Its thematic cross-links are the Gamma–harmonic resonance material, Hurwitz–Stieltjes calculus, and depth/cyclotomic chapters. The report contains a complete article and can be reviewed independently before any canonical chapter integration. All theorem and equation labels start with `dhr:` to reduce collisions.

The incoming procedure explicitly separates preservation, replay, analytic review, and canonical integration [4]. This delivery does not modify the GitHub repository, apply a patch, or promote its own results into the canonical manuscript. The separate `integration/INTEGRATION.md` records a suggested reading order and chapter mapping. The delivered source files, not this paragraph, are the proposed integration artifacts.

9 Further research questions

9.1 Higher resonances: several divergent moment entries

The first resonance $b - a - c = 2r - 2$ is special because only one particle can escape at logarithmic order while the lower ensemble retains a finite first moment. At $b - a - c = 2r - 2 - M$ with $M \geq 1$, a larger block of moments is divergent and the Gamma product can have higher pole order. Determine a canonical largest-index or multi-index subtraction, including all overlap terms, and compare its hard-cutoff polynomial in $\log N$ with its spectral Laurent coefficients. A successful extension must prove normal convergence after the full subtraction, not merely expand the Gamma product.

A precise initial target is the next resonance for $r = 2$, where one can explicitly compare the two divergent moments in the cutoff Hankel determinant with the two relevant gamma poles. This should clarify which derivatives of residue amplitudes enter beyond the first-order rule (3.12).

9.2 Repeated rational poles and the rank of harmonic data

The simple-pole hypothesis in Theorem 6.1 is essential to the proof. With repeated poles, moments contain $H^{(2)}_{N+1}, H^{(3)}_{N+1}, \ldots$ as well as H_{N+1}. Describe the matrix subspace spanned by these harmonic coefficients. Can its rank and support give a sharp *formal* upper bound on the polylogarithmic depth of the cutoff generator?

One concrete problem is to classify all rational weights for which this harmonic matrix pencil has rank one despite repeated poles. Such a classification could extend the current depth-two theorem without relying on generic determinant-degree estimates.

9.3 Closed all-rank coefficients and discrete orthogonal polynomials

The determinant formulas for $P_r(N)$ and $Q_r(N)$ are exact at every rank, but expression growth is substantial. Find product formulas, short recurrences, or a discrete-orthogonal-polynomial

interpretation for the rational coefficients in the canonical case. The factors of P_r displayed at ranks two through four suggest more structure than the determinant proof alone explains.

A useful target is a recurrence in r with rational coefficients in N that generates both P_r and Q_r and preserves the initial vanishing $S_{r,N} = 0$ for $N < r - 1$. A companion recurrence for $I_{r,q}$ could explain systematic weight drops in (7.13)–(7.16).

9.4 Mixed spectral and shape derivatives

Theorem 3.1 gives joint holomorphy in a, c, u. One can therefore differentiate in a, c as well as u, provided one keeps track of the moving relation $b_0 = a + c + 2r - 2$. Develop explicit harmonic/polygamma formulas for these tangential shape derivatives and compare them with derivatives taken at fixed b. The distinction changes the counterterm and is likely important for integration with the existing mixed-Dougall calculus.

A targeted problem is to derive a compact three-parameter Bell table whose rows express mixed generalized harmonic sums in terms of $\gamma_k(\alpha)$ and polygamma values at $a + r - 1, c + r - 1$. The analytic permission to differentiate is proved here; the most useful arithmetic basis for the resulting rows is not selected.

9.5 Rational grids beyond a common translate

The root filters in (6.21) and (6.22) handle one common residue class. For rational poles from several classes modulo a common denominator, the moment matrix can involve several independent shifted harmonic families. Determine whether its harmonic rank, rather than the number of summation variables, controls colored depth.

The first test case should use two residue classes with separate harmonic coefficients and an explicitly computed matrix pencil. Any resulting endpoint reduction must state its cyclotomic level and its precise relation system; ordinary-zeta closure should not be assumed from the common-integer-grid case.

9.6 Changing the Vandermonde exponent

The square Δ^2 is what permits Cauchy–Binet and a moment determinant. For $\Delta^{2\kappa}$ with integer $\kappa > 1$, one can still expand finitely in monomials, but the single-entry argument no longer applies automatically. Determine the first resonant decay, its subtraction amplitude, and a useful depth bound. A comparison with known discrete Selberg-type sums should precede any novelty claim in that direction.

9.7 Functional certificates and arithmetic conjectures

The current identities live in a tightly specified depth-two family. It is reasonable to ask whether translated or colored variants supply new functional relations useful for the manuscript's arithmetic conjectures. There is no result here showing that they settle S_6 or S_8. A productive target would be an exact functional certificate whose evaluation gives a previously missing relation, not another high-precision residual or an unsupported claim of period dependence.

9.8 Formalization and literature reconciliation

The rational determinant lemma, finite Cauchy–Binet identity, residue sum criterion, and terminating transform are natural first targets for formalization. The analytic normal-convergence argument should be formalized separately, with its compact-uniform first-moment estimate and Cauchy derivative transfer made explicit.

In parallel, a systematic literature audit should identify which forms of the Gauss–Vandermonde product and height-one generating functions already occur in discrete determinant and Euler-sum literature, then isolate the exact scope of the resonant cutoff and rank-one harmonic statements. The present report supplies proofs and reproducible identities; it does not turn an incomplete priority search into a novelty theorem.

A Reproducible computation and proof boundaries

A.1 What is in the package

The article consists of `article.tex`, the files under `sections/`, and `references.tex`. The standalone `article.pdf` is the compiled reading artifact. The mathematical software is

File	Purpose
`code/reductions.py`	Exact canonical cutoff, polylogarithm, and endpoint reducer
`code/verify_exact.py`	Finite rational/symbolic checks and formula export
`code/verify_numeric.py`	Separate floating-point diagnostics

The dependencies used in the recorded run are SymPy 1.14.0 and mpmath 1.3.0. Python 3.10 or later is suitable for the syntax in the scripts; the recorded run used Python 3.13. The scripts write their own output records, so replay should be performed on a working copy when the original delivery records need to be preserved.

Run from the package root:

```
python -m pip install -r requirements.txt
python code/verify_exact.py
python code/verify_numeric.py
lualatex -interaction=nonstopmode -halt-on-error article.tex
lualatex -interaction=nonstopmode -halt-on-error article.tex
```

A third LaTeX pass is harmless if a local build reports changed labels. No network access is needed by the mathematical scripts after the Python dependencies have been installed.

A.2 Exact assertions

The recorded exact suite passes 732 assertions. Grouping closely related checks gives the following accounting:

Check family	Assertions
Rational factorial determinant	18
Gamma product versus mixed moment determinant	16
Finite ordered sums, Cauchy–Binet, and finite moments	216
Rational partitions, residues, and cutoff constants	24
Polylogarithmic coefficients and removable origin terms	416
Nonconsecutive-pole residue and moment checks	32
Printed endpoint values and G_2	4
Intentionally corrupted functional formula rejected	1
Mixed harmonic coefficient rows	4
Printed Stieltjes row algebra	1
Total	732

These are finite exact checks. They detect arithmetic, index, and normalization mistakes in explicit cases; they do not replace the all-index proofs of Sections 2–6. The corruption control changes G_2 by z^2 and verifies that the identity check rejects it.

The file `results/exact_formulas.json` contains ranks $r = 1, 2, 3, 4$, their cutoff coefficients, rational spectral partitions, residues, generating functions of primitive orders $q = 0, 1, 2, 3$, and endpoint values for $q = 1, 2, 3$. In that file `L1`, `L2`, ... denote $\mathrm{Li}_1(z), \mathrm{Li}_2(z), \ldots$; `D1`, `D2`, ... denote $\mathrm{Li}_{1,1}(z, 1), \mathrm{Li}_{2,1}(z, 1), \ldots$. The coefficient expressions use exact rational numbers and SymPy syntax. They are not sampled numerical fits.

A.3 Numerical diagnostics

The recorded suite has 85 diagnostics, all at 100 decimal digits of *working precision*:

Diagnostic family	Cases
Generating functions and Euler primitives from direct lower moments	24
Ordinary Gauss product, nonintegral a, c	8
Convergent endpoints with an asymptotic tail	12
Spectral jets, $r = 1, \ldots, 4$, orders $0, \ldots, 4$, two α values	40
Halfpoint logarithm/dilogarithm value	1
Total	85

The largest recorded scaled discrepancy is approximately 2.75×10^{-32}. Individual tolerances and absolute as well as scaled discrepancies are stored in `results/numerical_checks.json`. It would be incorrect to describe all checks as agreements to 100 digits: working precision is not an error bound.

For disk evaluations, the left side is formed from directly accumulated lower moments and the right side from convergent polylogarithm series. For ordinary Gamma products, 1500 terms are used at a large positive excess. Endpoint diagnostics combine a 400-term direct head with a 30th-order Euler–Maclaurin expansion of the harmonic/rational tail, summed by Hurwitz-zeta values and order derivatives. These tail approximations are diagnostic, not interval-certified enclosures.

The spectral-jet left side uses an independent finite telescoping formula instead of the infinite

Gamma product. Namely,

$$\sum_{n=0}^{N}(n+1)_j\frac{\Gamma(n+1)}{\Gamma(n+2r+u)} = \frac{\Gamma(j+1)/\Gamma(2r-1+u)-\Gamma(N+j+2)/\Gamma(N+2r+u)}{2r-j-2+u}. \tag{A.1}$$

To prove it, subtract successive terms of $\Gamma(n+j+1)/\Gamma(n+2r-1+u)$; their difference is $(2r-j-2+u)\Gamma(n+j+1)/\Gamma(n+2r+u)$. At $j=2r-2, u=0$ the quotient has a removable singularity, which is evaluated by Taylor coefficients rather than by division of two rounded near-zero numbers. Transforming powers of n to rising factorials then gives the full finite moment determinant and its spectral derivatives. The diagnostic cutoff is $N=10^{40}$; it is used through this finite formula, not by enumerating that many summands.

The translated root-of-unity extension is proved algebraically in the article; the recorded 85-case numerical suite does not separately test that extension. This is part of the verification scope, not a gap filled by an unstated computation.

A.4 Proof, calculation, and compilation are different evidence

A clean LaTeX build establishes that the document compiles. Finite symbolic assertions establish the corresponding finite identities. Floating-point comparisons test implementations and motivate scrutiny. The all-order analytic and algebraic conclusions are justified by the proofs in the article. None of the three forms of computational evidence alone confers theorem-prover status or proves global novelty.

References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, canonical manuscript README, inspected 10 October 2026. Repository source anchor.

[2] ProveIt Contributors, *Mixed Gamma derivatives and exact harmonic tail identities*, `chapters/07-mixed-dougall-identities.tex`, in the canonical Polylogarithms manuscript, source commit `87c7db629552afbc2e803001526debe3430a31b0`. Pinned chapter source.

[3] ProveIt Contributors, *Mixed endpoints and all-weight cyclotomic product quotients*, `chapters/04-cyclotomic-quotients.tex`, same source commit. The first 240 source lines were inspected for conventions and the formal-versus-numerical distinction. Pinned chapter source.

[4] ProveIt Contributors, *Incoming reports*, `docs/incoming/README.md`, intake procedure and directory inventory inspected during this research session. The binary incoming archives were not audited here. Repository drop zone.

[5] NIST Digital Library of Mathematical Functions, §15.4, equation 15.4.20: Gauss hypergeometric summation. https://dlmf.nist.gov/15.4.E20.

[6] NIST Digital Library of Mathematical Functions, §5.11: asymptotic expansions and ratios of Gamma functions. https://dlmf.nist.gov/5.11.

[7] NIST Digital Library of Mathematical Functions, §5.12, equation 5.12.1: Euler's beta integral. https://dlmf.nist.gov/5.12.E1.

[8] NIST Digital Library of Mathematical Functions, §5.15: polygamma functions, shifts, and special values. https://dlmf.nist.gov/5.15.

[9] NIST Digital Library of Mathematical Functions, §25.11: Hurwitz zeta, parameter differentiation, special values, and the log-Gamma identity. https://dlmf.nist.gov/25.11.

[10] NIST Digital Library of Mathematical Functions, §25.12: polylogarithms and the dilogarithm reflection identity. https://dlmf.nist.gov/25.12.

[11] C. Krattenthaler, *Advanced determinant calculus*, Séminaire Lotharingien de Combinatoire **42** (1999), Article B42q, 67 pp.; arXiv:math/9902004. https://arxiv.org/abs/math/9902004. Cited as classical determinant-method background, not as a priority attribution for every formula in this report.